

# VIGNESH M

Full Stack Developer | AI Engineer | Node.js · React.js · GraphQL · AWS

Nagapattinam, Tamil Nadu, India | +91 6383527500 | vickypvm10@gmail.com [linkedin.com/in/vignesh-m-7107aa2b3](https://www.linkedin.com/in/vignesh-m-7107aa2b3)  
| [github.com/vignesh-ondrutech](https://github.com/vignesh-ondrutech)

## PROFESSIONAL SUMMARY

Results-driven Full Stack Developer with hands-on production experience building and deploying end-to-end web and AI-powered applications. Proficient in Node.js, React.js, GraphQL, and PostgreSQL for full stack development, with strong backend expertise in RESTful and GraphQL API design. Experienced in integrating AI capabilities using Claude AI, RAG pipelines, and LLM-based workflows into production systems. Deployed and managed applications on AWS EC2 and S3 with a focus on performance, scalability, and security. Passionate about delivering clean, maintainable code and building software that solves real-world problems.

## TECHNICAL SKILLS

**Frontend:** React.js — Component architecture, REST API integration, responsive UI development

**Backend:** Node.js — RESTful APIs, GraphQL API design, server-side logic and middleware

**Database:** PostgreSQL — Relational schema design, query optimization, indexing strategies

**API:** GraphQL — Schema design, resolvers, and query optimization for scalable data APIs

**Cloud & DevOps:** AWS (EC2, S3, IAM) — Application deployment, static asset storage, access control

**Language:** JavaScript — Full stack development across Node.js and React.js ecosystems

**Version Control:** Git / GitHub — Source control, branching, pull requests, and collaborative development

**AI Engineering:** RAG pipelines, LLM integration, vector embeddings, semantic search, OCR pipelines

**AI Tools:** Claude AI, Anthropic SDK, LangChain, FAISS / Chroma Vector DB, Prompt Engineering

## WORK EXPERIENCE

**Full Stack Developer & AI Engineer** | **Ondru Technologies** — Nagapattinam, Tamil Nadu *January 2024 – Present*

- Architected and shipped a production-grade RAG pipeline using Python, FastAPI, Vector DB, and Claude AI, powering a live Android application on the Google Play Store.
- Deployed and maintained production applications on AWS EC2, utilizing S3 for static asset storage and IAM for role-based access control, ensuring 99%+ uptime.
- Developed an OCR-based question ingestion pipeline that automatically processes and indexes over 10,000 TNPSC exam questions from PDFs with high accuracy.
- Engineered scalable GraphQL APIs using Node.js and PostgreSQL for a financial jewellery savings platform, supporting complex multi-role business logic.
- Optimized database queries and implemented indexing strategies, reducing API response times by 35%.
- Secured applications using JWT authentication and Role-Based Access Control (RBAC) for multi-user, multi-role environments.
- Collaborated with cross-functional teams to design, test, and deliver production-ready full stack features on schedule.

## PROJECTS

**TNPSC AI Learning App** [github.com/vignesh-ondrutech/ondru-tnpsc-api](https://github.com/vignesh-ondrutech/ondru-tnpsc-api)

*Live on Google Play Store | Python · FastAPI · Vector DB · Claude AI · AWS EC2*

- Designed and implemented an end-to-end RAG pipeline using Python, FastAPI, and a Vector Database for semantic retrieval of exam content.
- Integrated Claude AI (LLM) to power intelligent Q&A functionality, reducing manual content curation effort by 60%.

- Built an OCR pipeline to automatically extract and process over 10,000 exam questions from PDFs, improving content ingestion accuracy and speed.
- Achieved sub-200ms query response times using vector embedding-based semantic search for real-time user interactions.
- Deployed RESTful APIs on AWS EC2, serving the live Android application with 99%+ uptime and secure configurations.

#### **VKS Jewels Financial Backend** [github.com/Ondrutech-Technologies/OneProJewel](https://github.com/Ondrutech-Technologies/OneProJewel)

*Node.js · GraphQL · PostgreSQL · JWT · RBAC*

- Engineered a production-grade backend using Node.js, GraphQL, and PostgreSQL for a jewellery savings and financial management platform.
- Designed a normalized relational database schema to manage customer accounts, gold accumulation tracking, and full transaction history.
- Optimized complex SQL queries and implemented indexing, improving data retrieval speed by 35%.
- Implemented JWT authentication and RBAC to support secure, multi-role access control across the platform.

#### **Alliance Bank V3 Web Application** [github.com/Ondrutech-Technologies/v3template](https://github.com/Ondrutech-Technologies/v3template)

*React.js · REST API · Component Library*

- Developed a modular, scalable banking web application using React.js with a focus on reusability and performance.
- Built a reusable component library that reduced future feature development time by 40%.
- Integrated REST APIs for real-time data rendering, improving overall application responsiveness and user experience.

## **EDUCATION**

---

**B.Sc in Computer Science** — Bharathidasan University, Trichy

*Currently Pursuing | Expected Graduation: 2027*

**Diploma in Computer Science Engineering** — Tamil Nadu State Board

*Graduated: 2024*